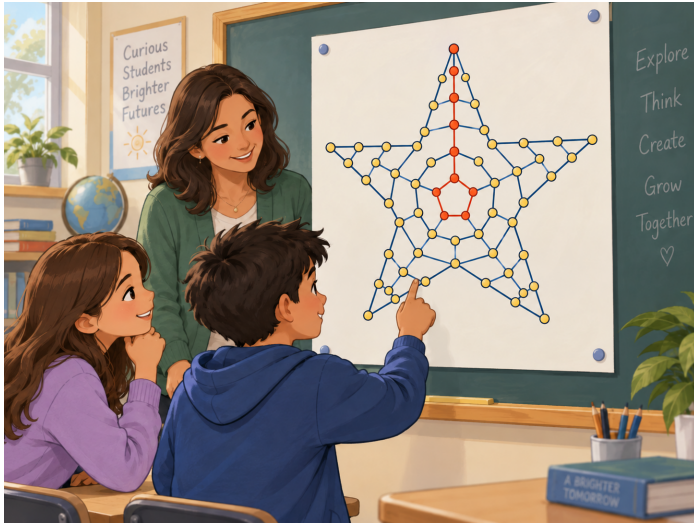


★ Meet The Great American Star

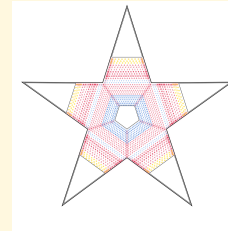
Observe the structure. Make a prediction. Explain your thinking.

Name: _____

Date: _____



The big idea



50 starting points

50 ending points

359,277,250
possible paths

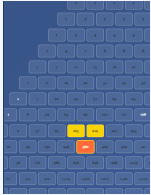
1 American star

Mathematical art created for America 250.

The complete star is made of 5 matching sections.

In this activity, we study one section.

The same path-counting pattern repeats in the other four sections.



A modified Pascal's Triangle

The path structure is based on a modified form of Pascal's Triangle.

Choices branch and recombine, so the number of possible routes grows quickly.

Each of the five star sections follows the same basic path-counting pattern.

Your first prediction

We will now study ONE section. Which starting position do you think creates the MOST possible paths?

- ☐ Near the edge ☐ In the middle ☐ At a corner ☐ Not sure yet

Why do you think so?

What do you notice?

Write one pattern, symmetry, or repeated structure you notice in the star.





Compare Two Starting Positions

Both starts can reach all 10 endings. Now compare how many routes are possible.

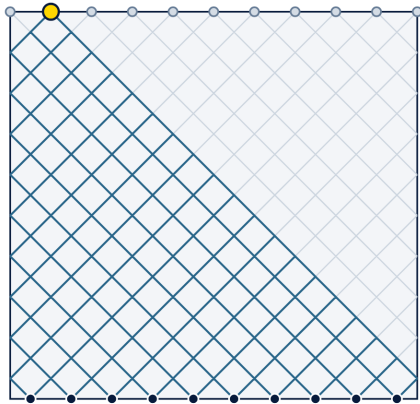


Simplified illustration — representative routes only

● Selected start — Valid branches from this start

Near-Edge Start

11 starting points · top



Same 10 ending points · bottom

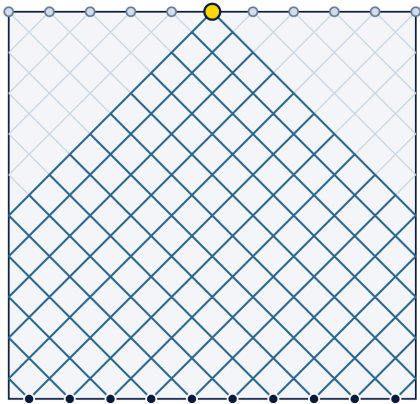
Can reach all 10 ending points, but reaching endings on the far side requires many steps to keep moving in that direction, leaving fewer ways to vary the route.

5,546,105

possible paths through one section

Middle Start

11 starting points · top



Same 10 ending points · bottom

Can reach all 10 ending points, with more freedom to vary left/right choices while still arriving at the same endings.

8,097,610

possible paths through one section

Same ending points. Different number of routes.

Both starting points can reach all 10 ending points.

The middle can reach them through more different routes because it has more freedom to vary its left/right choices while still arriving at the same endings.

Think about the difference

1. Which starting position has more paths through one section?

2. Both starts can reach all 10 ending points. Why can their total path counts still be different?

3. In your own words, what does "more freedom to vary the route" mean?

Challenge

Near-edge: 5,546,105 paths Middle: 8,097,610 paths

How many MORE paths does the middle have? _____



First calculate. Then use the visual to check your reasoning.



A corner contributes 4,194,280 paths from Section A and 4,194,280 from Section B.

$$4,194,280 + 4,194,280 =$$

The middle has 8,097,610 paths. How many more paths does the shared corner have?

- 8,097,610 =



How can a corner have fewer paths inside one section, but more paths than the middle in the complete star?

There are 359,277,250 possible paths in the complete star. One of those paths can become your path.

What could make your path feel like yours?

"This one is mine."

